

PRISMA 2020 Flow Diagram

Systematic Review: Foundation Models for Retail Demand Forecasting

IDENTIFICATION

Records identified through
database searching
(n = 150)

5 databases, 30+ queries
(Semantic Scholar, Google Scholar,
arXiv, Scopus, Web of Science)

Additional records from
other sources
(n = 0)

GIFT-Eval leaderboard,
citation chasing, grey literature

SCREENING

Records after duplicates removed
(n = 120)

Duplicates removed
(n = 30)

Titles/abstracts screened
(n = 120)

Records excluded
(n = 50)

Not demand/retail forecasting,
no quantitative results,
duplicate/superseded

ELIGIBILITY

Full-text articles assessed
for eligibility
(n = 70)

Full-text articles excluded
(n = 32)

No retail dataset (n=12)
No FM evaluation (n=10)
Duplicate results (n=10)

Studies included
(n = 38: 32 papers
+ 6 own experiment blocks)

Source B: own experiments
(n = 6 blocks, 45 rows)

Gap-filling experiments:
FM + baselines on M5,
Favorita, Rohlik v2 (\$5)

INCLUDED

Extraction rows
(n = 185)

Category Breakdown

- A: Foundation model papers (n=17, 58 rows)
- B: Benchmark papers (n=4, 40 rows)
- C: Retail ML/DL papers (n=8, 31 rows)
- D: Cross-domain papers (n=3, 3 rows)
- F: fev-bench entries (n=1, 8 rows)
- OWN: Gap-filling experiments (n=6, 45 rows)

Extraction: 185 rows
across 38 studies
in extraction_schema.csv